

Physics-Based Animation



Some Slides/Images adapted from Marschner and Shirley

Physics-Based Animation

Agenda

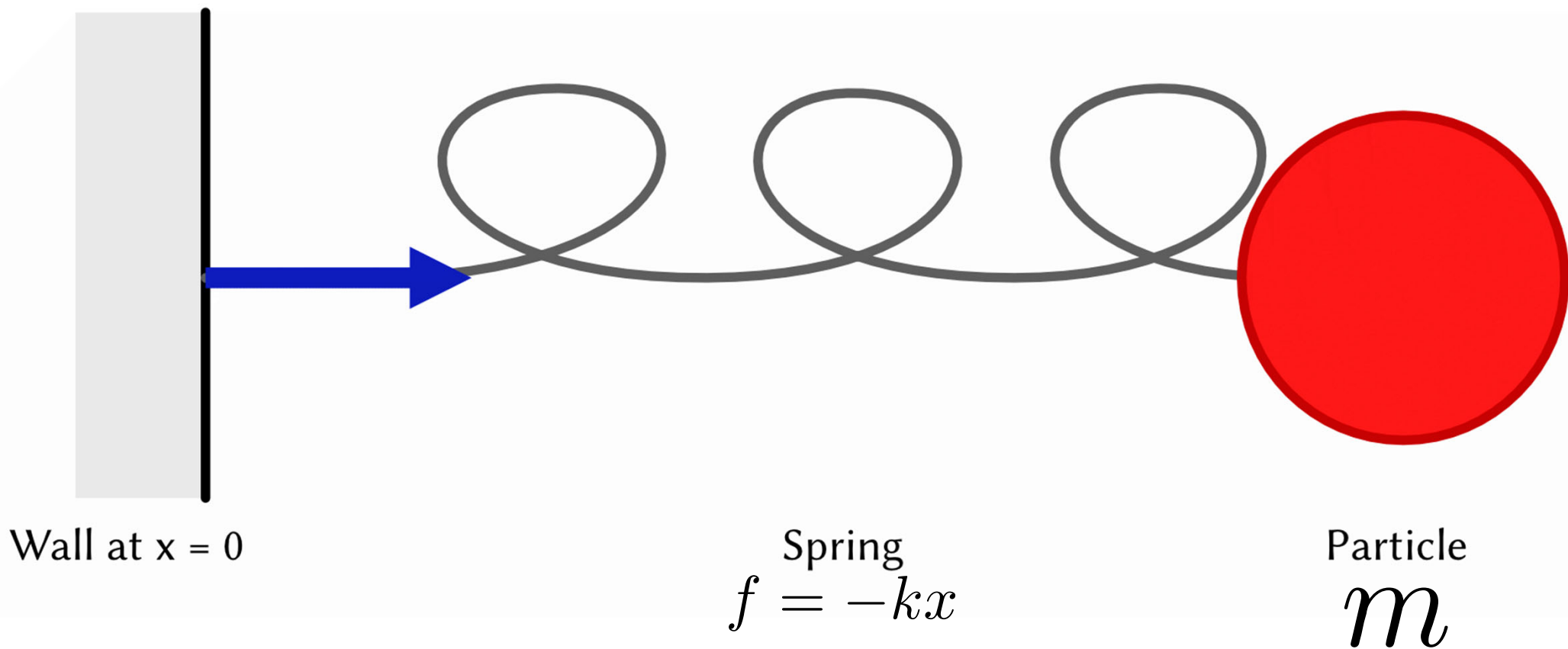
- Newton's Laws of Motion
- The Mass-Spring System
- Implicit Integration via Optimization
- A Local-Global Solver for Fast-Mass Springs

Newton's Laws

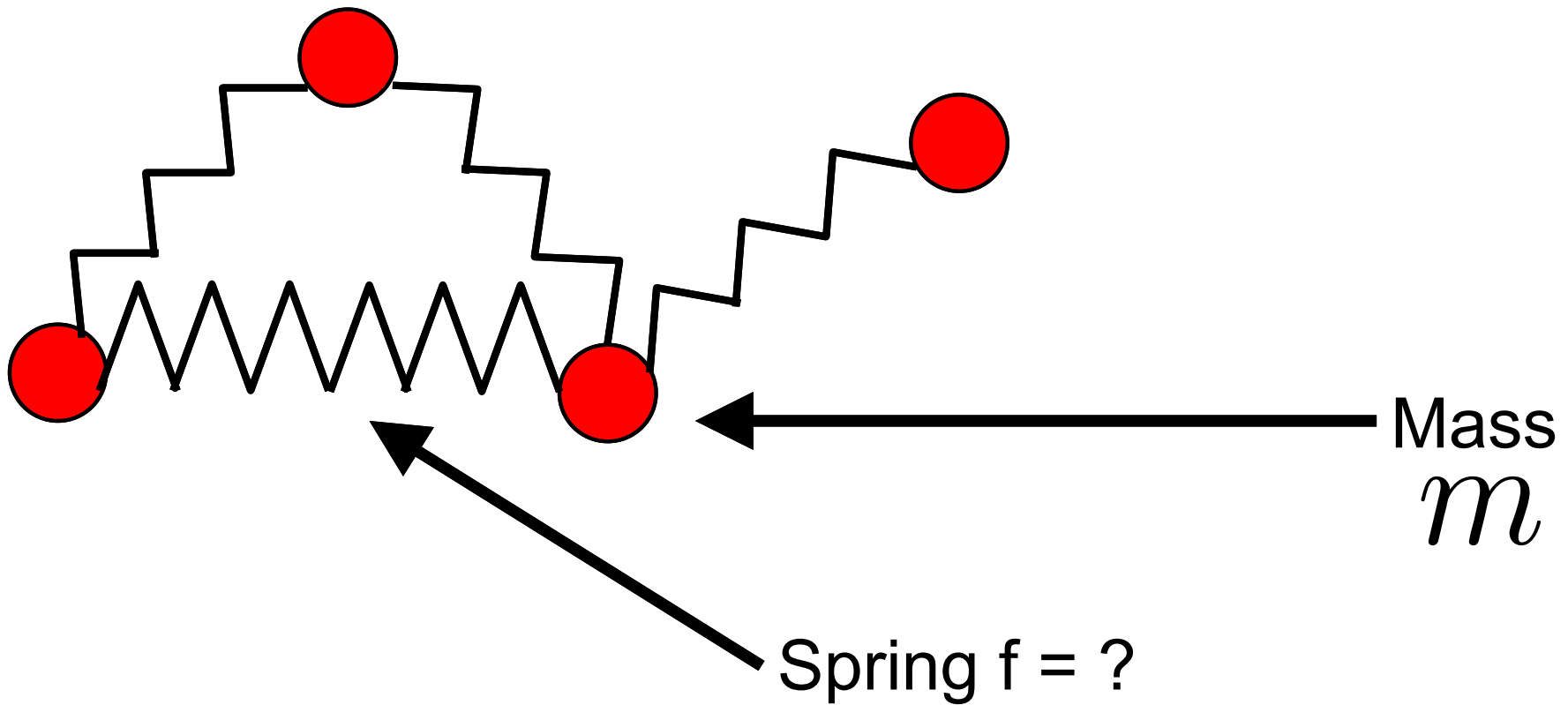
1. Every object will remain at rest or in uniform motion in a straight line unless compelled to change its state by the action of an external force.
2. The force acting on an object is equal to the time rate-of-change of the momentum.
3. For every action there is an equal and opposite reaction.

Newton's Second Law

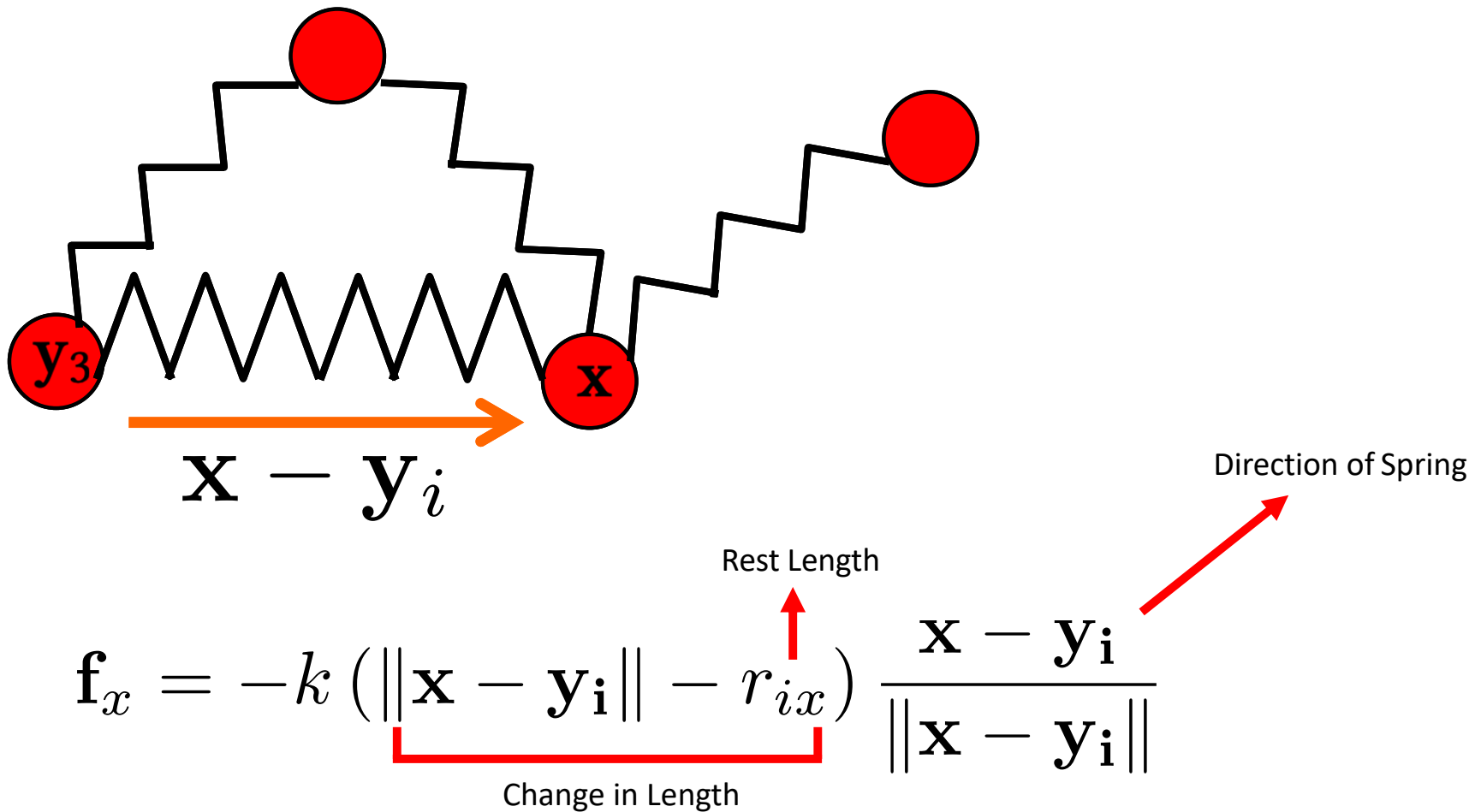
$$\underset{\text{Mass}}{m} \overset{\text{Acceleration}}{a} = \underset{\text{force}}{f}$$



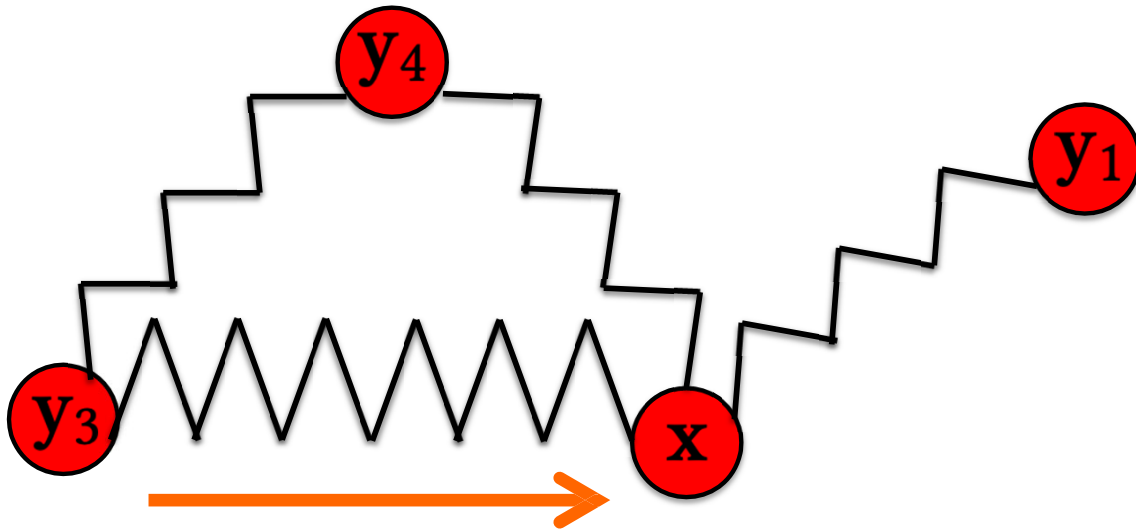
The Mass-Spring System



The Mass-Spring System

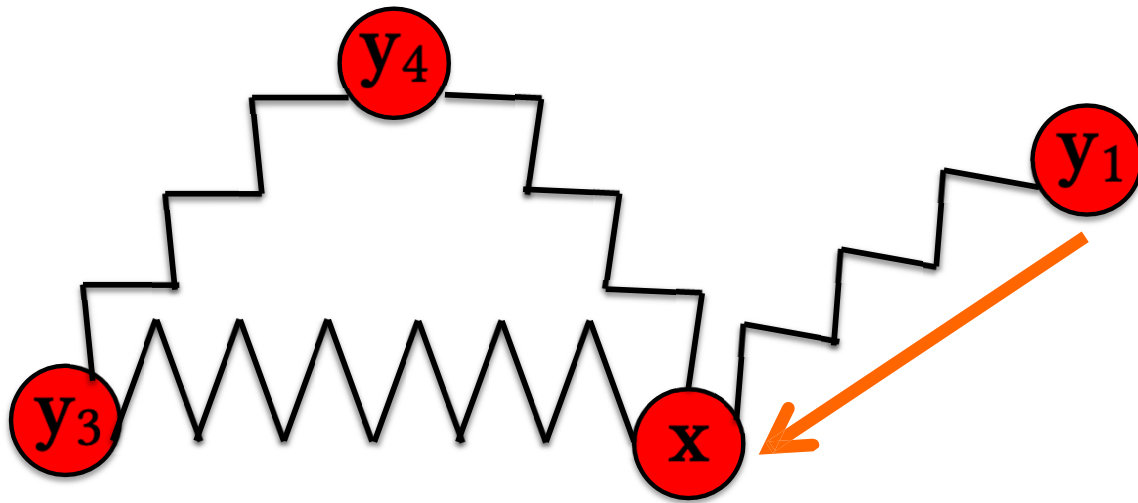


Newton's Second Law for Each Particle



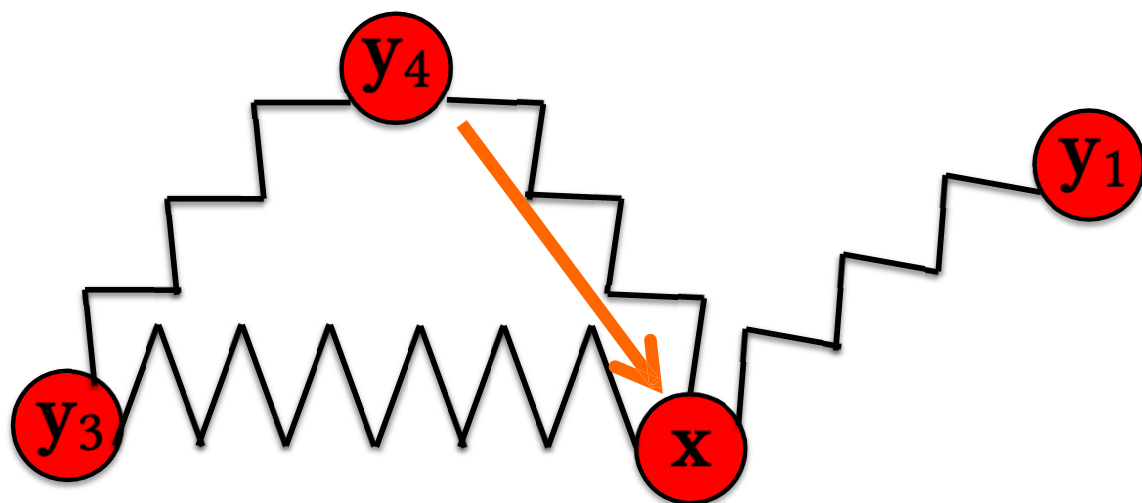
$$m_x \mathbf{a}_x = \sum_i \mathbf{f}_x(\mathbf{y}_i)$$

Newton's Second Law for Each Particle



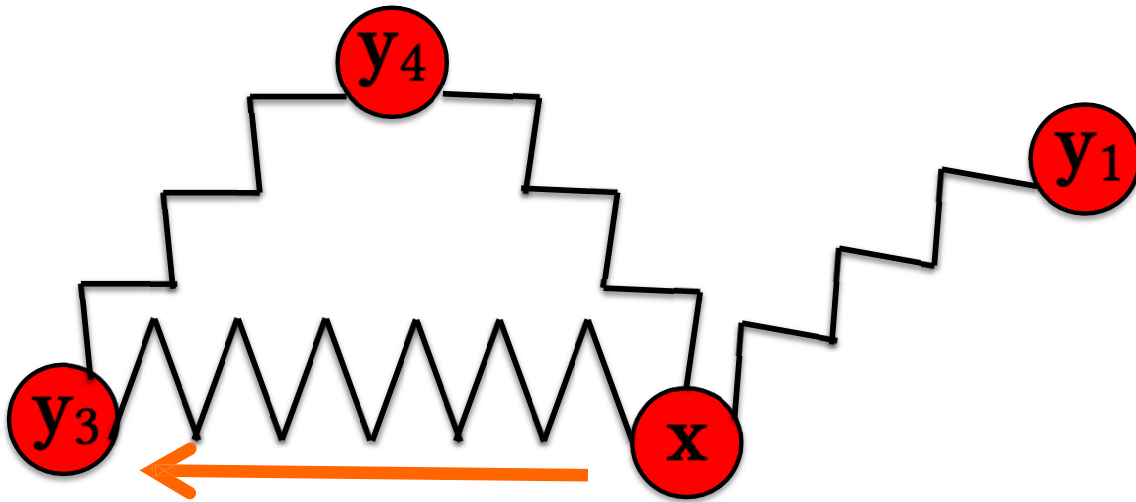
$$m_x \mathbf{a}_x = \sum_i \mathbf{f}_x(\mathbf{y}_i)$$

Newton's Second Law for Each Particle



$$m_x \mathbf{a}_x = \sum_i \mathbf{f}_x(\mathbf{y}_i)$$

Newton's Second Law for Each Particle



$$m_x \mathbf{a}_x = \sum_i \mathbf{f}_x(\mathbf{y}_i)$$

One equation for each object/particle.

We will solve them all together.





Cloth

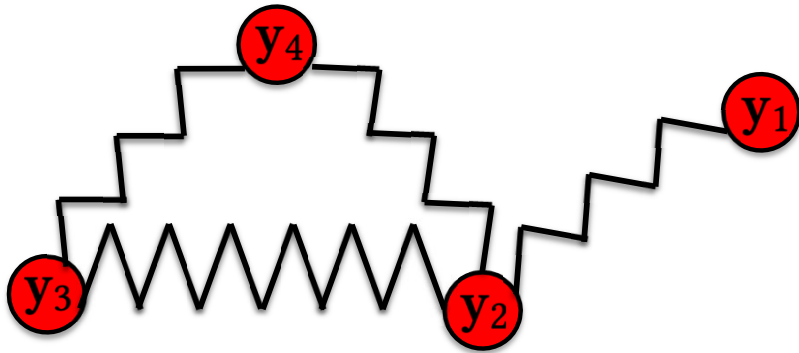
SIMIT GPU

15,630 Triangles

7,988 Verts

14 FPS

Newton's Second Law: System of Equations



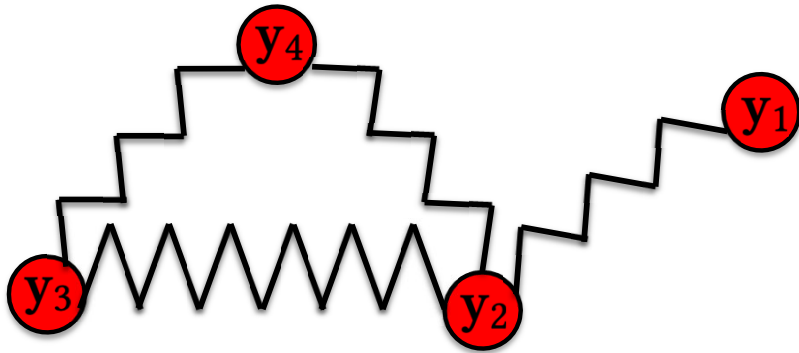
$$m_1 \mathbf{a}_1 = \sum_i \mathbf{f}_1(\mathbf{y}_i)$$

$$m_2 \mathbf{a}_2 = \sum_i \mathbf{f}_2(\mathbf{y}_i)$$

$$m_3 \mathbf{a}_3 = \sum_i \mathbf{f}_3(\mathbf{y}_i)$$

$$m_4 \mathbf{a}_4 = \sum_i \mathbf{f}_4(\mathbf{y}_i)$$

Newton's Second Law: System of Equations



$$\begin{pmatrix} m_1 \cdot I & 0 & 0 & 0 \\ 0 & m_2 \cdot I & 0 & 0 \\ 0 & 0 & m_3 \cdot I & 0 \\ 0 & 0 & 0 & m_4 \cdot I \end{pmatrix} \begin{pmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \mathbf{a}_3 \\ \mathbf{a}_4 \end{pmatrix} = \begin{pmatrix} \mathbf{f}_1 \\ \mathbf{f}_2 \\ \mathbf{f}_3 \\ \mathbf{f}_4 \end{pmatrix}$$

Mass Matrix $\mathbf{a}(t)$ $\mathbf{f}(t)$

Time Integration

$$M \mathbf{a}(t) = \mathbf{f}(\mathbf{y}(t))$$

$$M \frac{d^2 \mathbf{y}(t)}{dt^2} = \mathbf{f}(\mathbf{y}(t))$$

Time Integration

$$M \frac{d^2 \mathbf{y}(t)}{dt^2} = \mathbf{f}(\mathbf{y}(t))$$

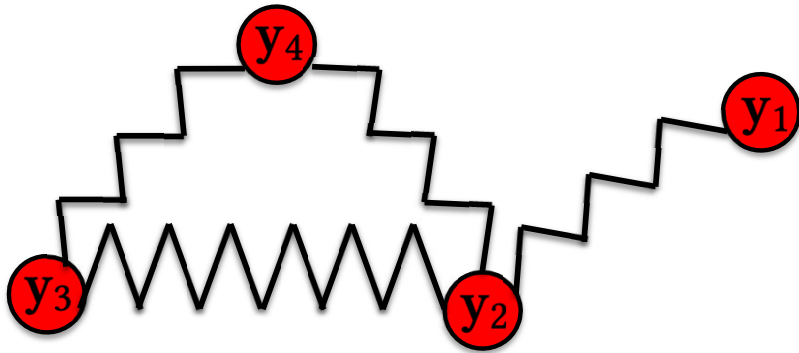
Use Finite Differences!

$$\frac{d^2 \mathbf{y}(t)}{dt^2} \approx \frac{\mathbf{y}^{t+1} - 2\mathbf{y}^t + \mathbf{y}^{t-1}}{\Delta t^2}$$

https://en.wikipedia.org/wiki/Finite_difference#Higher-order_differences

https://en.wikipedia.org/wiki/Finite_difference_coefficient#Central_finite_difference

Time Integration



Need to Discretize!

$$M \frac{d^2 \mathbf{y}(t)}{dt^2} = \mathbf{f}(\mathbf{y}(t))$$

Use Finite Differences!

$$\frac{d^2 \mathbf{y}(t)}{dt^2} \approx \frac{\mathbf{y}^{t+1} - 2\mathbf{y}^t + \mathbf{y}^{t-1}}{\Delta t^2}$$

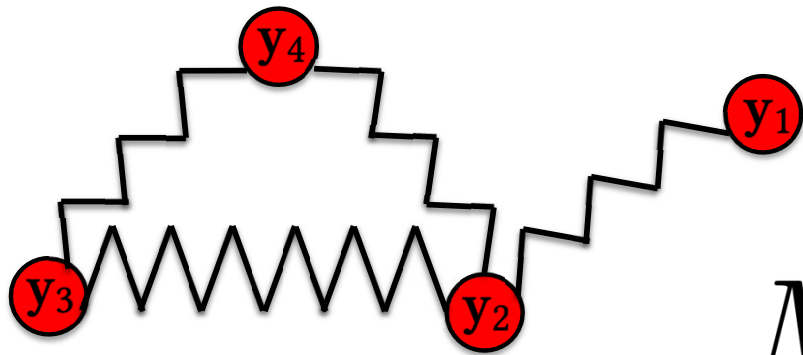
Time Integration: Explicit vs. Implicit

$$M \frac{d^2 \mathbf{y}(t)}{dt^2} = \mathbf{f}(\mathbf{y}(t))$$

Explicit: $\mathbf{y}_{t+dt} = \mathbf{g}(\mathbf{y}_t)$. Future state $\mathbf{y}_{t+dt}$ is an explicit equation of current state $\mathbf{y}_t$ and dt .

Implicit: $\mathbf{h}(\mathbf{y}_t, \mathbf{y}_{t+dt}) = 0$. Future state $\mathbf{y}_{t+dt}$ is an implicit equation.

Implicit Time Integration



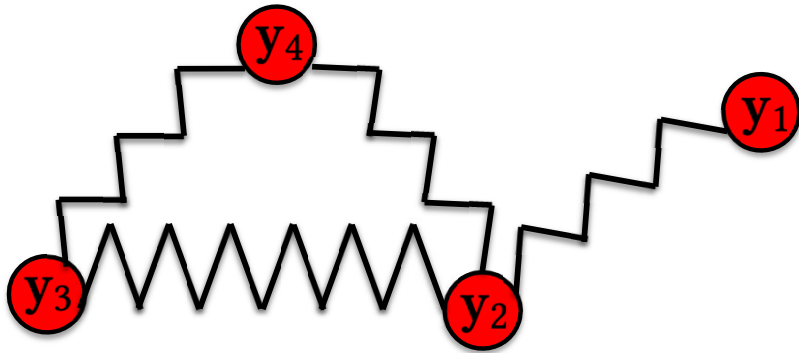
$$M \frac{d^2 \mathbf{y}}{dt^2} (t) = \mathbf{f} (\mathbf{y}^{t+1})$$

A red arrow points to the $\mathbf{y}^{t+1}$ term in the equation.

Use Finite Differences!

$$\frac{d^2 \mathbf{y}(t)}{dt^2} \approx \frac{\mathbf{y}^{t+1} - 2\mathbf{y}^t + \mathbf{y}^{t-1}}{\Delta t^2}$$

Implicit Time Integration



$$M \left(\frac{y^{t+1} - 2y^t + y^{t-1}}{\Delta t^2} \right) = f(y^{t+1})$$

$$M y^{t+1} = M (2y^t - y^{t-1}) + \Delta t^2 f(y^{t+1})$$

Goal: Solve for y^{t+1}

Implicit Integration as Optimization

Rather than directly solve:

$$M\mathbf{y}^{t+1} - M(2\mathbf{y}^t - \mathbf{y}^{t-1}) - \Delta t^2 \mathbf{f}(\mathbf{y}^{t+1}) = 0$$

spring forces +
external (e.g. gravity) forces



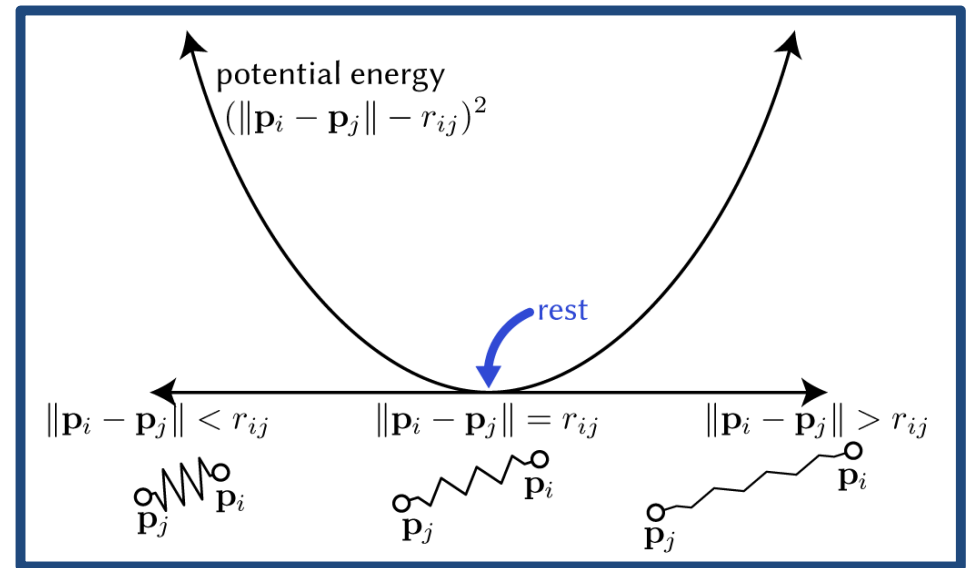
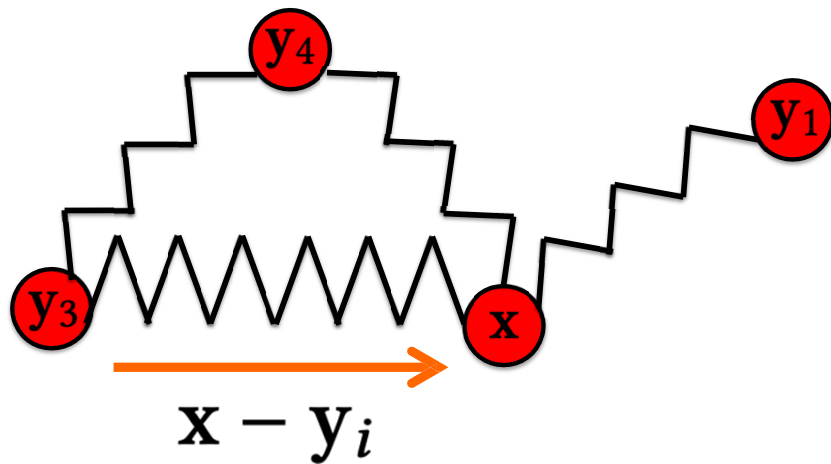
We can view the mass-force equations as an energy function $\mathbf{E}(\mathbf{q})$ whose gradient vanishes as above:

$$\nabla_{\mathbf{q}} E(\mathbf{y}^{t+1}) = 0$$

Turning integration into an optimization problem:

$$\mathbf{y}^{t+1} = \arg \min_{\mathbf{q}} E(\mathbf{q})$$

Mass-Spring Potential Energy



$$V(\mathbf{y}_i, \mathbf{x}) = \frac{1}{2} k (\underbrace{\|\mathbf{y}_i - \mathbf{x}\| - r_{ix}}_{\text{Change in Length}})^2$$

Rest Length

$-\nabla V$ $\rightarrow$ $\mathbf{f}_x(\mathbf{y}_i) = -k(\|\mathbf{x} - \mathbf{y}_i\| - r_{ix}) \frac{\mathbf{x} - \mathbf{y}_i}{\|\mathbf{x} - \mathbf{y}_i\|}$

Energy

$$\mathbf{y}^{t+1} = \underset{\mathbf{y}}{\operatorname{argmin}} \underbrace{\left(\sum_{ij} \frac{1}{2} k (\|\mathbf{y}_i - \mathbf{y}_j\| - r_{ij})^2 \right) - \frac{\Delta t^2}{2} \left(\sum_i m_i \left(\frac{\mathbf{y}_i - 2\mathbf{y}_i^t + \mathbf{y}_i^{t-1}}{\Delta t^2} \right)^2 \right) - \left(\sum_i \mathbf{y}_i^\top \mathbf{f}_i^{\text{ext}} \right)}_{E(\mathbf{y})}$$

Potential energy

$$V(\mathbf{y}_i, \mathbf{x}) = \frac{1}{2} k (\|\mathbf{y}_i - \mathbf{x}\| - r_{ix})^2$$

Energy

$$\mathbf{y}^{t+1} = \underset{\mathbf{y}}{\operatorname{argmin}} \underbrace{\left(\sum_{ij} \frac{1}{2} k (\|\mathbf{y}_i - \mathbf{y}_j\| - r_{ij})^2 \right) - \frac{\Delta t^2}{2} \left(\sum_i m_i \left(\frac{\mathbf{y}_i - 2\mathbf{y}_i^t + \mathbf{y}_i^{t-1}}{\Delta t^2} \right)^2 \right) - \left(\sum_i \mathbf{y}_i^\top \mathbf{f}_i^{\text{ext}} \right)}_{E(\mathbf{y})}$$

Potential energy

$$V(\mathbf{y}_i, \mathbf{x}) = \frac{1}{2} k (\|\mathbf{y}_i - \mathbf{x}\| - r_{ix})^2$$

0.5*ma²

Kinetic energy-like

$$\mathbf{a}_i^t = \ddot{\mathbf{y}}_i^t = \frac{d^2 \mathbf{y}_i(t)}{dt^2} \approx \frac{\mathbf{y}_i^{t+1} - 2\mathbf{y}_i^t + \mathbf{y}_i^{t-1}}{\Delta t^2}$$

Energy

$$\mathbf{y}^{t+1} = \underset{\mathbf{y}}{\operatorname{argmin}} \underbrace{\left(\sum_{ij} \frac{1}{2} k (\|\mathbf{y}_i - \mathbf{y}_j\| - r_{ij})^2 \right) - \frac{\Delta t^2}{2} \left(\sum_i m_i \left(\frac{\mathbf{y}_i - 2\mathbf{y}_i^t + \mathbf{y}_i^{t-1}}{\Delta t^2} \right)^2 \right)}_{E(\mathbf{y})} - \boxed{\left(\sum_i \mathbf{y}_i^\top \mathbf{f}_i^{\text{ext}} \right)}$$

External forces

Potential energy force

$$V(\mathbf{y}_i, \mathbf{x}) = \frac{1}{2} k (\|\mathbf{y}_i - \mathbf{x}\| - r_{ix})^2$$

0.5*ma²
Kinetic energy-like

$$\mathbf{a}_i^t = \ddot{\mathbf{y}}_i^t = \frac{d^2 \mathbf{y}_i(t)}{dt^2} \approx \frac{\mathbf{y}_i^{t+1} - 2\mathbf{y}_i^t + \mathbf{y}_i^{t-1}}{\Delta t^2}$$

Energy

Construct a function E , such that its minimizer is a simulation solution $\nabla E = \mathbf{f} - \mathbf{m}\mathbf{a}$

$$\mathbf{y}^{t+1} = \operatorname{argmin}_{\mathbf{y}} \underbrace{\left(\sum_{ij} \frac{1}{2} k (\|\mathbf{y}_i - \mathbf{y}_j\| - r_{ij})^2 \right) - \frac{\Delta t^2}{2} \left(\sum_i m_i \left(\frac{\mathbf{y}_i - 2\mathbf{y}_i^t + \mathbf{y}_i^{t-1}}{\Delta t^2} \right)^2 \right) - \left(\sum_i \mathbf{y}_i^\top \mathbf{f}_i^{\text{ext}} \right)}_{E(\mathbf{y})}$$

...verify that $\nabla E = \mathbf{0}$ is indeed the force equation below.

$$M \left(\frac{\mathbf{y}^{t+1} - 2\mathbf{y}^t + \mathbf{y}^{t-1}}{\Delta t^2} \right) = \mathbf{f}(\mathbf{y}^{t+1})$$

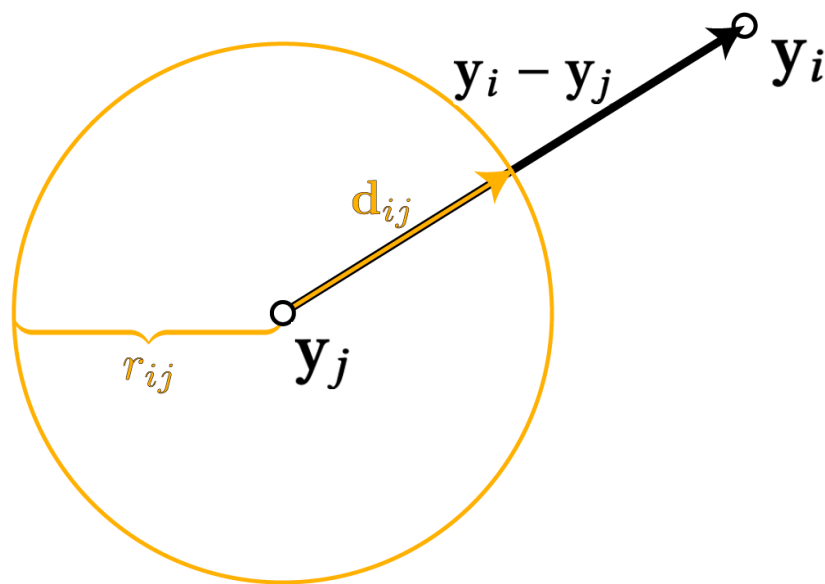
Energy

$$\mathbf{y}^{t+1} = \underset{\mathbf{y}}{\operatorname{argmin}} \underbrace{\left(\sum_{ij} \frac{1}{2} k (\|\mathbf{y}_i - \mathbf{y}_j\| - r_{ij})^2 \right) - \frac{\Delta t^2}{2} \left(\sum_i m_i \left(\frac{\mathbf{y}_i - 2\mathbf{y}_i^t + \mathbf{y}_i^{t-1}}{\Delta t^2} \right)^2 \right) - \left(\sum_i \mathbf{y}_i^\top \mathbf{f}_i^{\text{ext}} \right)}_{E(\mathbf{y})}$$

Non linear :(

Observation!

$$(\|\mathbf{y}_i - \mathbf{y}_j\| - r_{ij})^2 = \min_{\mathbf{d}_{ij} \in \mathbb{R}^3, \|\mathbf{d}\|=r_{ij}} \|(\mathbf{y}_i - \mathbf{y}_j) - \mathbf{d}_{ij}\|^2$$



Observation!

$$(\|\mathbf{y}_i - \mathbf{y}_j\| - r_{ij})^2 = \min_{\mathbf{d}_{ij} \in \mathbb{R}^3, \|\mathbf{d}\|=r_{ij}} \|(\mathbf{y}_i - \mathbf{y}_j) - \mathbf{d}_{ij}\|^2$$

$$\mathbf{y}^{t+1} = \operatorname{argmin}_{\mathbf{y}} \underbrace{\left(\sum_{ij} \frac{1}{2} k \|(\mathbf{y}_i - \mathbf{y}_j) - \mathbf{d}_{ij}\|^2 \right)}_{E_2(\mathbf{y})} - \underbrace{\frac{\Delta t^2}{2} \left(\sum_i m_i \left(\frac{\mathbf{y}_i - 2\mathbf{y}_i^t + \mathbf{y}_i^{t-1}}{\Delta t^2} \right)^2 \right)}_{\tilde{E}(\mathbf{y})} - \underbrace{\left(\sum_i \mathbf{y}_i^\top \mathbf{f}_i^{\text{ext}} \right)}_{E_I(\mathbf{y})}$$

Quadratic!

Local-Global Solvers for Mass-Spring Systems

$$\mathbf{E}_1 (\mathbf{y}^{t+1}) = \frac{1}{2} (\mathbf{y}^{t+1})^T M \mathbf{y}^{t+1} - (\mathbf{y}^{t+1})^T M \mathbf{b}$$

where $\mathbf{b} = 2\mathbf{y}^t - \mathbf{y}^{t-1}$

$$E_2 = \sum_{ij} \frac{k}{2} \left(\|\mathbf{y}_i - \mathbf{y}_j\|^2 - 2(\mathbf{y}_i - \mathbf{y}_j)^T \mathbf{d}_{ij} + \mathbf{d}_{ij}^T \mathbf{d}_{ij} \right)$$

Both energies are quadratic now. This will let us build a fast algorithm

We will do this using block coordinate descent. First optimize over one set of variables (the d's) then the second set (the y's) Rinse and repeat!

Local-Global Solvers for Mass-Spring Systems

WHILE Not done

For Each Spring

 Local Optimization

 Global Optimization

END

Now we can start defining these steps for mass-springs

The Local Step

Hold $\mathbf{y}$ constant and optimize each spring vector $\mathbf{d}$

$$\arg \min_{\mathbf{d}_{ij}, |\mathbf{d}_{ij}|=r_{ij}} \sum_{ij} \frac{k}{2} \|\mathbf{y}_i - \mathbf{y}_j\|^2 - 2(\mathbf{y}_i - \mathbf{y}_j)^T \mathbf{d}_{ij} + \mathbf{d}_{ij}^T \mathbf{d}_{ij}$$

Rotate $\mathbf{d}$'s to align with current $\mathbf{y}$'s.

$$E_{ij} = \arg \min_{\mathbf{d}_{ij}, |\mathbf{d}_{ij}|=r_{ij}} \frac{k}{2} \underbrace{\|\mathbf{y}_i - \mathbf{y}_j - \mathbf{d}_{ij}\|^2}_{\text{No sum anymore!}}$$

The Global Step

Minimizing wrt to $\mathbf{y}$ requires us to find

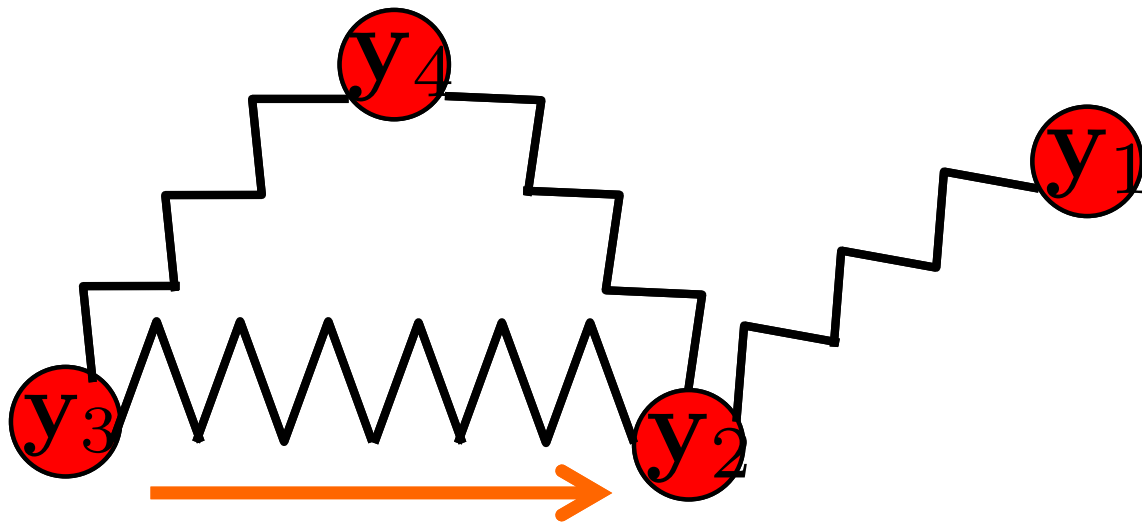
$$\mathbf{y}^{t+1} \text{ s.t. } \nabla_{\mathbf{y}}(E_1(\mathbf{y}) + \Delta t^2 E_2(\mathbf{y}, \mathbf{d}_{ij})) = \mathbf{0}$$

Recall $\mathbf{E}_1(\mathbf{y}) = \frac{1}{2} \mathbf{y}^T M \mathbf{y} - \mathbf{y}^T M \mathbf{b}$

$$\nabla \mathbf{E}_1 = M \mathbf{y} - M \mathbf{b} \qquad \mathbf{b} = 2\mathbf{y}^t - \mathbf{y}^{t-1}$$

$$E_2 = \sum_{ij} \frac{k}{2} \|\mathbf{y}_i - \mathbf{y}_j\|^2 - 2(\mathbf{y}_i - \mathbf{y}_j)^T \mathbf{d}_{ij} + \mathbf{d}_{ij}^T \mathbf{d}_{ij}$$

The Global Step



$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}$$

$$\Delta \mathbf{y} = \begin{pmatrix} I & -I & 0 & 0 \\ 0 & I & -I & 0 \\ 0 & I & 0 & -I \\ 0 & -I & I & -I \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}$$

G

Each row is a spring

The Global Step

Using this we can rewrite the second energy as

$$E_2 = \frac{k}{2} (\mathbf{y} G^T G \mathbf{y} - 2 \mathbf{y}^T G^T \mathbf{d} + \mathbf{d}^T \mathbf{d})$$

So the gradient becomes

$$\nabla E_2 = k G^T G \mathbf{y} - k \mathbf{G}^T \mathbf{d}$$

And the total global step finds $\mathbf{y}$ so that

$$\nabla(E_1 + \Delta t^2 E_2) = (M + \Delta t^2 k G^T G) \mathbf{y} - (M \mathbf{b} - \Delta t^2 k \mathbf{G}^T \mathbf{d}) = 0$$

$$\mathbf{d} = \begin{pmatrix} \mathbf{d}_{12} \\ \mathbf{d}_{23} \\ \mathbf{d}_{24} \\ \mathbf{d}_{34} \end{pmatrix}$$

The Global Step

And the total global step finds $\mathbf{y}$ so that

$$\nabla(E_1 + \Delta t^2 E_2) = (M + \Delta t^2 k G^T G) \mathbf{y} - (M \mathbf{b} - \Delta t^2 k \mathbf{G}^T \mathbf{d}) = 0$$

or

$$(M + \Delta t^2 k G^T G) \mathbf{y} = (M \mathbf{b} - \Delta t^2 k \mathbf{G}^T \mathbf{d})$$

You can solve this linear system using the Cholesky Solver in Eigen

Local-Global Solvers for Mass-Spring Systems

WHILE Not done

//Local Steps

For Each Spring

$$E_{ij} = \arg \min_{\mathbf{d}_{ij}, |\mathbf{d}_{ij}|=r_{ij}} \frac{k}{2} \|\mathbf{y}_i - \mathbf{y}_j - \mathbf{d}_{ij}\|^2$$

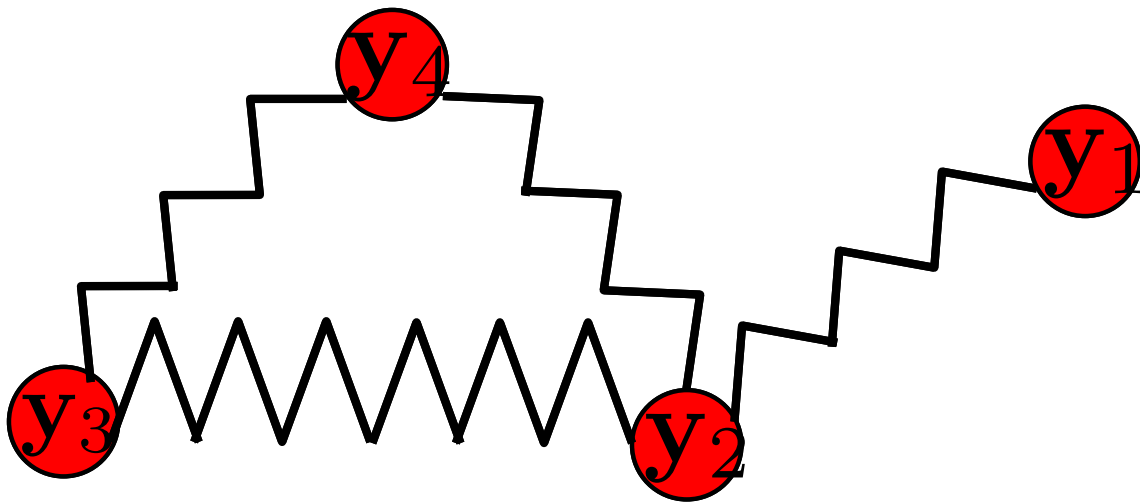
//Global Step

Solve $(M + \Delta t^2 k G^T G) \mathbf{y} = (M \mathbf{b} + \Delta t^2 k \mathbf{G}^T \mathbf{d})$

END



Fixed Points

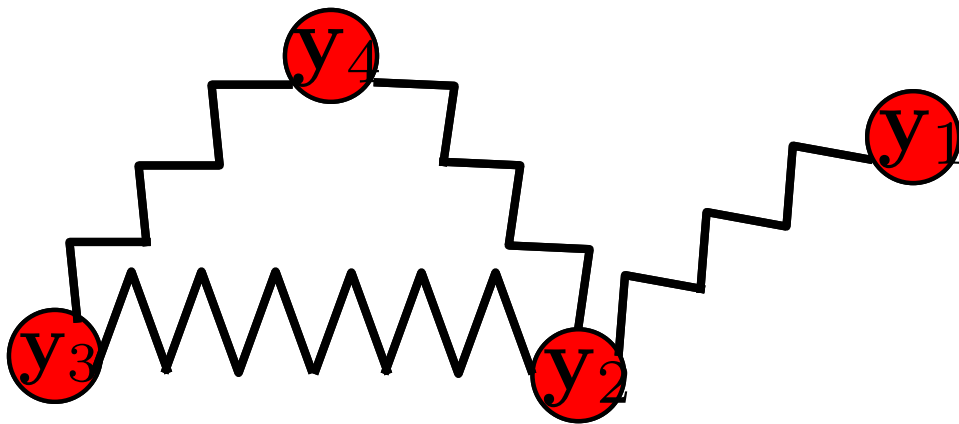


$$\mathbf{y} = \begin{pmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \mathbf{y}_3 \\ \mathbf{y}_4 \end{pmatrix}$$

Let's say we never want $\mathbf{y}_3$ to move:

i.e $\mathbf{y}_3 = \mathbf{c}$ forever and always

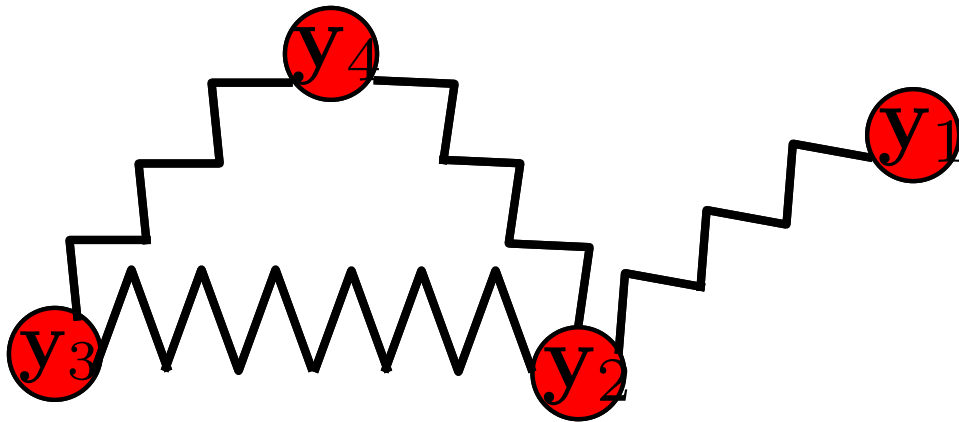
Fixed Points via Projection



$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}$$

$$\mathbf{y} = \underbrace{\begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & 0 \\ 0 & 0 & I \end{pmatrix}}_{\mathbf{P}} \underbrace{\begin{pmatrix} y_1 \\ y_2 \\ y_4 \end{pmatrix}}_{\tilde{\mathbf{y}}} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ \mathbf{c} \\ 0 \end{pmatrix}}_{\mathbf{c}}$$

Fixed Points via Projection



$$(M + \Delta t^2 k G^T G) \mathbf{y} = (M \mathbf{b} - \Delta t^2 k \mathbf{G}^T \mathbf{d})$$

Substituting $\mathbf{y} = P \tilde{\mathbf{y}} + \mathbf{c}$ in $A \mathbf{y} = \mathbf{f}$

Too many rows now ... $A P \tilde{\mathbf{y}} = \mathbf{f} - A \mathbf{c}$

$$P^T A P \tilde{\mathbf{y}} = P^T (\mathbf{f} - A \mathbf{c}) \quad \text{...Rebuild } \mathbf{y}$$